

TheFatCat Whitepaper

Decentralized Trading-Tax Routing Protocol with Adaptive Damping and Seniority Weighting

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TheFatCat Protocol Core Contributors · Based on Project Plan v0.5 and verified against working tree with 74 suites / 538 tests

This document contains no representations of anticipated return on investment or yield commitments

Abstract

Common decentralized dividend and trading-tax designs enforce globally shared reward assets, pass fee flows directly through with little temporal damping, and allocate proceeds by capital balance alone without seniority weighting. These are characteristics of upstream templates compared here, not universal claims across all fee-routing tokens.

TheFatCat composes the following mechanisms into an integrated protocol architecture:

1. **The Belly Hydrodynamic Reservoir:** Permissionless clock progression every 8 hours minimum. At nominal cadence, it disburses $\frac{1}{21} \approx 4.7619\%$ of unreserved quote balance per interval, yielding a zero-inflow model half-life of 4.735 days;
2. **O(1) Closed-Form Seniority Ledger:** Decomposes cohorts algebraically to aggregate seniority notches 1 through 22 without iterating over positions. Complete redemption scales with execution batches spanned, bounded via `claimThrough`;
3. **Sovereign Diet Architecture:** Autonomous asset selection per position. Ignoring observation limits and integer dust, nominal quote budgets depend on global weight share; asset choice affects execution, market value, issuer risk, and external incentives;
4. **Dual-Liability Accounting & Conservative Rounding:** WBNB default and fallback quote asset, batch accounting, verified delivery balance assertions, and integer floor truncation maintaining tested solvency bounds.

1 Introduction & Paradigm Shift

1.1 Structural Dilemmas in Direct Pass-Through Models

In decentralized automated market maker (AMM) ecosystems, trading-tax tokens typically route swap fees directly into uniform reward tokens for token holders. Upstream templates exhibit three structural constraints:

1. **Globally Shared Reward Selection:** All eligible holders share a single asset or fixed basket without per-position preference;
2. **Absence of Temporal Buffering:** Fee inflows drop sharply when trading volume contracts, transmitting volatility instantly;
3. **Absence of Seniority Input:** Allocation depends strictly on snapshot capital without rewarding duration. Transient capital can dilute historical holders during high-volume spikes.

1.2 The Reservoir Model: Damping Flow into Sticky Inventory

TheFatCat transforms the pass-through pipe into a self-damped physical reservoir—**The Belly**. Fee tokens are converted into unreserved quote capital and released via deterministic clock intervals. Transient volatility is converted into cross-interval inventory.

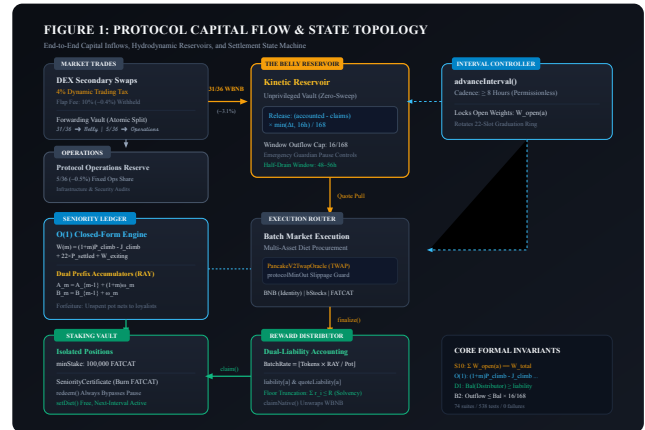


Figure 1: TheFatCat Protocol Capital Flow Topology and State Machine Architecture

2 The Belly Hydrodynamics & Time-Proration

2.1 First-Order Exponential Decay and Steady-State Level

The Belly disburses a fraction of **unreserved quote balance** (accounted - outstandingClaims), not gross contract balance.

At base cadence, intervals require $\Delta t_{\min} = 8$ hours, benchmarked against one week ($T_{\text{week}} = 168$ hours). The nominal release coefficient α is:

$$\alpha = \frac{\Delta t_{\min}}{T_{\text{week}}} = \frac{8}{168} = \frac{1}{21} \approx 4.7619\%$$

Let B_k be unreserved quote balance before interval k . Assuming zero new inflow, regular 8-hour ticks, full execution, and negligible dust:

$$B_{k+1} = B_k \cdot (1 - \alpha) = B_k \cdot \left(\frac{20}{21}\right)$$

By mathematical induction:

$$B(k) = B_0 \cdot \left(\frac{20}{21}\right)^k$$

With a constant fee inflow of I every 8 hours, steady-state balance satisfies $B_{eq} \cdot \alpha = I$:

$$B_{eq} = \frac{I}{\alpha} = 21 \cdot I$$

Ideal Steady State: Under identical assumptions, steady-state balance is $21I$, equivalent to seven days of continuous inflows. Actual balance varies with batch arrivals, keeper delays, liabilities, and rounding.

2.2 Half-Life Analytical Derivation

Setting $B(k) = \frac{1}{2}B_0$, the half-life interval count $k_{\frac{1}{2}}$ satisfies:

$$\left(\frac{20}{21}\right)^{k_{\frac{1}{2}}} = \frac{1}{2} \Rightarrow k_{\frac{1}{2}} = \frac{\ln(0.5)}{\ln(\frac{20}{21})} \approx 14.207 \text{ intervals}$$

Converting to physical elapsed time $t_{\frac{1}{2}}$:

$$t_{\frac{1}{2}} = k_{\frac{1}{2}} \times 8 \text{ hours} \approx 113.65 \text{ hours} \approx 4.735 \text{ days}$$

Elapsed Time	Interval k	Remaining %	Exact Value
Day 1 (24h)	3	$\left(\frac{20}{21}\right)^3$	86.38%
Day 3 (72h)	9	$\left(\frac{20}{21}\right)^9$	64.46%
Day 7 (168h)	21	$\left(\frac{20}{21}\right)^{21}$	35.89%
Day 14 (336h)	42	$\left(\frac{20}{21}\right)^{42}$	12.88%
Day 30 (720h)	90	$\left(\frac{20}{21}\right)^{90}$	1.24%

Reserve capital is not mathematically exhausted after 7 days of zero volume; after 30 days, approximately 1.24% of the initial balance remains.

2.3 Dynamic Time-Proration and Discrete Rate Limits

1. Single-Release Span Ceiling (CAP_SPAN):

In `IntervalController.sol`, maximum prorated time is capped at $CAP_SPAN = 2 \times MIN_INTERVAL = 16$ hours:

$$\text{Allocation} = \left\lfloor (\text{accounted} - \text{claims}) \cdot \frac{\min(\Delta t, CAP_SPAN)}{T_{\text{week}}} \right\rfloor$$

2. Vault Outflow Rate Limit (windowAllowance):

In `Belly.sol`, immutable parameters restrict maximum window outflow:

$$\text{MaxOutflow} = \left\lfloor \text{Balance} \cdot \frac{16}{168} \right\rfloor \approx 9.5238\%$$

Even under compromised spenders, continuous 50% drawdown equivalent is:

$$n_{\frac{1}{2}}^{\text{eq}} = \frac{\ln(0.5)}{\ln(\frac{152}{168})} \approx 6.92569 \text{ windows} \approx 55.4055 \text{ hours}$$

Discrete execution requires seven full window allowances (49.6295% remaining after 7th). Spanning 6 intervals, it takes 48 hours if initial allowance is usable, and 56 hours if waiting for window rollover.

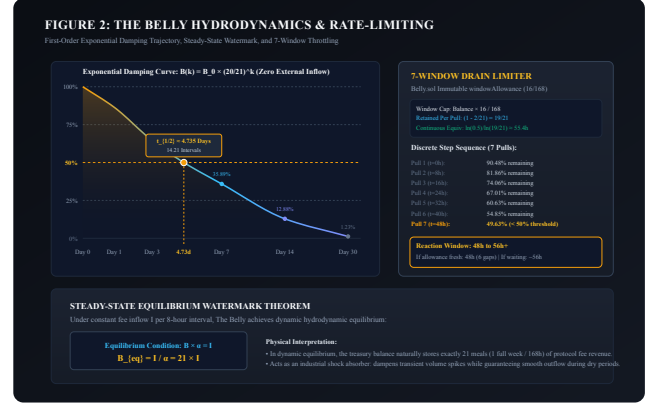


Figure 2: The Belly Hydrodynamic Decay Curve and Discrete Rate-Limiter Defensive Model

3 Seniority Ledger & Scalable Topology

3.1 Independent Positions and State Isolation

Each FATCAT stake creates an isolated position across two contracts:

$$\text{Position}_i = \{\text{owner}, \text{principal}, \text{diet}, \text{activeFrom}, \text{endInterval}\}$$

Anti-merging: Positions cannot be merged, preventing seniority whitewashing; Next-interval Activation: Stakes in interval m become active at $\text{activeFrom} = m + 1$, neutralizing flash-loan dividend exploits.

3.2 $O(1)$ Closed-Form Seniority Aggregation

Position i with principal p_i accrues seniority notch $c_{i(m)} \in [1, 22]$:

$$c_{i(m)} = 1 + \min(m - j_i, 21), \quad j_i = \text{activeFrom}_i$$

Positions divide into:

- Climbing Cohort:** $m - j_i < 21$;
- Settled Cohort:** $m - j_i \geq 21$.

Summing over climbing positions:

$$\sum_{i \in \text{Climb}} p_i(1 + m - j_i) = (1 + m) \sum p_i - \sum (p_i j_i)$$

The contract maintains two global scalars: $P_{\text{climb}} = \sum_{i \in \text{Climb}} p_i$
 $J_{\text{climb}} = \sum_{i \in \text{Climb}} (p_i \cdot j_i)$

Total active weight evaluation is strictly $O(1)$:

$$W(m) = (1 + m)P_{\text{climb}} - J_{\text{climb}} + 22P_{\text{settled}} + W_{\text{exiting}}$$

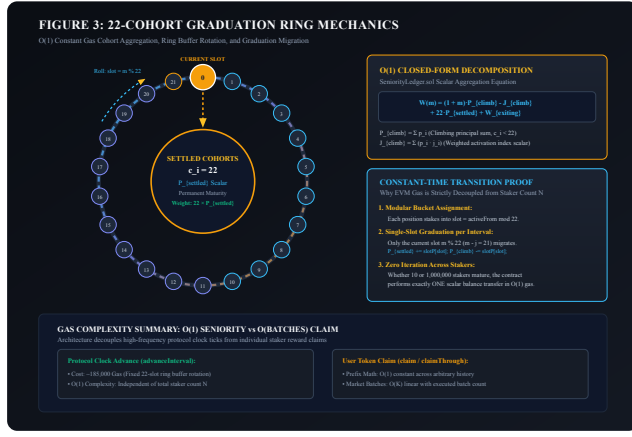


Figure 3: 22-Cohort Graduation Ring Mechanics

3.3 Dual-Accumulator Formulation

Per-interval yield per unit weight for asset a scaled by RAY = 10^{27} :

$$\omega_{a,m} = \left\lfloor \frac{\text{Share}_{a,m} \cdot \text{RAY}}{W_{open}(a, m)} \right\rfloor$$

Prefix accumulators:

$$A_{a,m} = A_{a,m-1} + (1 + m) \cdot \omega_{a,m}$$

$$B_{a,m} = B_{a,m-1} + \omega_{a,m}$$

Climbing position quote entitlement:

$$\text{Entitlement}(j, m) = \left\lfloor \frac{p \cdot (\Delta A_a - j \cdot \Delta B_a)}{\text{RAY}} \right\rfloor$$

Settled position quote entitlement:

$$\text{Entitlement}_{\text{settled}} = \left\lfloor \frac{22 \cdot p \cdot \Delta B_a}{\text{RAY}} \right\rfloor$$

Dual prefix subtraction yields $O(1)$ entitlement arithmetic per clean interval. Complete token redemption requires $O(K)$ batch iteration, boundable via `claimThrough`.

3.4 Mid-Interval Forfeiture Netting

Positions can redeem principal without lockup penalties (`redeem()`), forfeiting the active interval's meal:

$$\text{Forfeited}_a = \left\lfloor \text{Allocation}_a \cdot \frac{w_{\text{exit}}}{W_{open}(a)} \right\rfloor$$

Forfeited funds remain permanently in asset a 's pot, reducing denominator for subsequent batch execution: $\text{ClaimablePot}_a = \text{Pot}_a - \text{Forfeited}_a$, increasing `BatchRate` for remaining participants.

4 Allocation Neutrality & Seniority Scenarios

4.1 Nominal Quote Allocation Under Ideal Conditions

Under security property S10, $\sum_{a \in \text{MENU}} W_{open}(a) = W_{\text{total}} \cdot \text{Asset } a \text{ allocation is:}$

$$Q_a = Q_{\text{total}} \cdot \frac{W_{open}(a)}{W_{\text{total}}}$$

Position i 's entitlement is:

$$\text{Share}_{i,a} = Q_a \cdot \frac{w_i}{W_{open}(a)} = Q_{\text{total}} \cdot \frac{w_i}{W_{\text{total}}}$$

Denominators cancel in real numbers without probation caps.

Position quote budget is $\frac{w_i}{W_{\text{total}}}$, independent of cohort size.

4.2 Seniority Relative Advantage Analysis

Comparison	Multiplier	Economic Context
Max Mature vs Notch 1	22.0x	Temporary disparity, parity in 21 ticks
Max Mature vs Pool Average	$\frac{22}{\bar{c}}$	Distribution-dependent, $\bar{c} \in [1, 22]$
Mature in Stagnant Pool	1.0x	All mature, premium converges to 1

With principal-weighted average seniority $\bar{c} \in [1, 22]$, relative advantage $\frac{22}{\bar{c}}$ spans 1.0x to 22.0x. Uniform distribution ($\bar{c} = 11.5$) yields 1.913x as a representative scenario, not a mathematical upper bound.

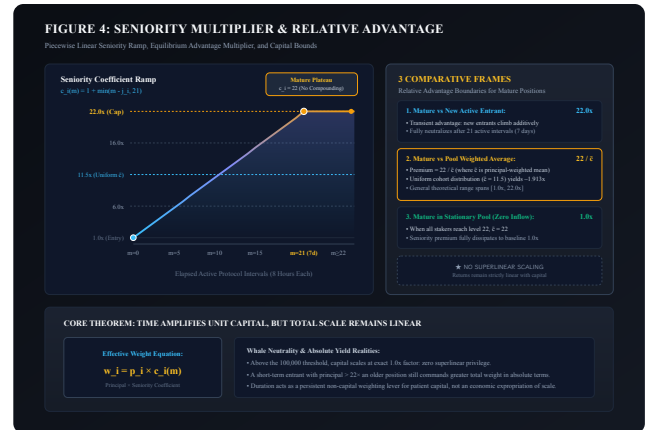


Figure 4: Seniority Multiplier, Settled Premium Curve, and Linear Principal Bounds

4.3 Minimum Stake and Seniority Certificate Roadmap

- Immutable Minimum Stake:** `minStake` set to 100_000 FATCAT (0.01% supply), no setter post-deployment;
- Strict Linear Scaling:** Weight scales strictly linearly with capital beyond threshold;
- Seniority Exit & Certificate Roadmap:** Planned as a subsequent roadmap module; stakers who fully redeem principal may burn FATCAT to $0x \dots dEaD$ to permanently imprint their achieved seniority notch (1–22) into a 100% on-chain SVG ERC-721 Seniority Certificate. Complete parameters and utility will be revealed ahead of rollout.

5 Execution & Solvency Accounting

5.1 Dual-Liability Accounting System

RewardDistributor maintains two separate liability ledgers: Token Liabilities (`liability[asset]`): **D18{tok} dimension; Quote Liabilities (`quoteLiability[asset]`): D18{quote} dimension.**

Batch exchange rate:

$$\text{BatchRate} = \left\lfloor \frac{\text{TokensReceived} \cdot \text{RAY}}{\text{ClaimablePot}} \right\rfloor$$

Stalled swaps exceed timeout (`maxPendingAge`, 3 days default), enabling permissionless `fallbackFinalize()`.

5.2 On-Chain TWAP and Execution Bounds

1. **FATCAT Pair:** PancakeV2TwapOracle reads cumulative price for TWAP;
2. **bStock Pairs:** Time-weighted liquidity verification;
3. **Slippage Protection:** Router verifies:

$$\text{minOut} = \left\lfloor \frac{\text{expectedOut} \cdot (10000 - \text{deviationBps})}{10000} \right\rfloor$$

5.3 Conservative Rounding and Solvency Invariants

For each batch with $\sum_i x_i \leq S$:

$$\text{Rate} = \left\lfloor \frac{R \cdot \text{RAY}}{S} \right\rfloor, \quad r_i = \left\lfloor \frac{x_i \cdot \text{Rate}}{\text{RAY}} \right\rfloor$$

Guarantees:

$$\sum_i r_i \leq R$$

Tested property D1 invariants: `balanceOf(Distributor, a) ≥ liability[a]`

`balanceOf(Distributor, quote) ≥  $\sum_a \text{quoteLiability}[a]$`

6 Security Posture & Dependencies

6.1 Restricted Vault Privileges

1. **The Belly:** Has immutable governor and guardian roles (governance 2-of-3 Gnosis Safe). Ops Safe receives 5/36 revenue atomically with zero protocol privileges. No admin sweep, rescue, or withdrawal. Spender activation is one-time with timelock;
2. **StakingVault:** `redeem()` intentionally ignores pause. Pause blocks only new deposits.

6.2 Tax Processor External Ownership

Tax processor is retained by Flap factory post-graduation. Tax lasts for 100 years post-graduation (becoming permanently zero thereafter), and after a 1-year anti-farmer period taxes only the primary liquidity pool. Protocol provides `watch-processor.sh` to poll state. Direct custody risk affects future fees, not principal.

6.3 Default Diet and RWA Risk Isolation

1. **WBNB Default and Fallback:** Identity route requires no market swap, eliminating bStock RWA risk;
2. **bStock Credit Isolation:** 5% probation cap restricts exposure;
3. **FATCAT Listing:** Requires 3-day governance timelock post-graduation.

7 Verification & Disclaimers

7.1 Reproducible Test Evidence

Baseline: **74 test suites / 538 tests / 0 failures** (FOUNDRY_PROFILE=local forge test). Property tests do not equate to formal verification.

7.2 Deterministic Scope and Non-Goals

1. **Not a Fixed-Income Product:** Zero APY commitments;
2. **Not a Lockup:** Principal withdrawable in any block;
3. **Zero Superlinear Bias:** Capital scales strictly linearly;
4. **Not a Pension:** Half-life 4.735 days under zero volume; 1.24% balance after 30 days.

8 Appendix A: Notation & Dimensions Index

Symbol	Semantic Definition	Domain Bound	EVM Dimension Scale
m	Global interval index	$\mathbb{N}^+$	Dimensionless (uint256)
j_i	Active from interval	$m_{\text{stake}} + 1$	Dimensionless (uint256)
p_i	Staked principal	$\geq 100,000$	D18{FAT-CAT}
$c_{i(m)}$	Seniority notch	Integer [1, 22]	Dimensionless scalar
$w_{i(m)}$	Effective weight	$p_i \cdot c_{i(m)}$	D18{weight}
$W(m)$	Total active weight	$\sum w_{i(m)}$	D18{weight}
W_{open}	Asset open denominator	$\sum W_{\text{open}} = W$	D18{weight}
P_{climb}	Climbing principal sum	$\sum p_i$	D18{FAT-CAT}
J_{climb}	Climbing weighted index	$\sum (p_i j_i)$	D18{FAT-CAT}
P_{settled}	Settled principal sum	$\sum p_i$	D18{FAT-CAT}
W_{exiting}	Residual exit weight	$\sum w_i$	D18{weight}
RAY	Accumulator scale	Constant 10^{27}	D27 scale
$\omega_{a,m}$	Yield per unit weight	Yield scalar	D27 (uint256)
$A_{a,m}$	Accumulator A	$\sum (1 + k)\omega$	D27 prefix sum
$B_{a,m}$	Accumulator B	$\sum \omega$	D27 prefix sum
α	Belly release fraction	$\frac{8}{168} = \frac{1}{21}$	Dimensionless ($\approx 4.76\%$)

B_k	Belly available balance	Unreserved quote	D18{quote} (WBNB)
BatchRateBatch	conversion rate	Fixed ex-change rate	D27 ratio
liability	Token liability	≥ 0	D18{tok}
quoteLiab	Quote asset liability	≥ 0	D18{quote}

9 Appendix B: Solidity Rounding and Solvency Bounds

9.1 Integer Floor Division

$$\forall a, b \in \mathbb{N}^+, \left\lfloor \frac{a}{b} \right\rfloor = \frac{a - (a \bmod b)}{b} \leq \frac{a}{b}$$

Rounding error $\varepsilon = \frac{a \bmod b}{b}$ satisfies $0 \leq \varepsilon < 1$.

9.2 Non-negative Dust Theorem

Theorem: Given token purchase R , pot S , and $\sum_{i=1}^N x_i \leq S$:

$$\sum_{i=1}^N r_i \leq R$$

Proof:

1. Rate definition: $\text{Rate} = \left\lfloor \frac{R \cdot \text{RAY}}{S} \right\rfloor$
2. Position claim: $r_i = \left\lfloor \frac{x_i \cdot \text{Rate}}{\text{RAY}} \right\rfloor \leq \frac{x_i \cdot \text{Rate}}{\text{RAY}}$
3. Summing across N positions:

$$\sum_{i=1}^N r_i \leq \frac{\text{Rate}}{\text{RAY}} \sum_{i=1}^N x_i$$

4. Substituting $\sum x_i \leq S$ and $\text{Rate} \leq \frac{R \cdot \text{RAY}}{S}$:

$$\sum_{i=1}^N r_i \leq \frac{\text{Rate} \cdot S}{\text{RAY}} \leq \frac{\left(\frac{R \cdot \text{RAY}}{S} \right) \cdot S}{\text{RAY}} = R \quad \blacksquare$$

Corollary: Residual dust $\Delta_{\text{dust}} = R - \sum_{i=1}^N r_i \geq 0$ settles permanently in distributor.

9.3 Solvency Preconditions

1. **Standard Token Behavior:** No fee-on-transfer, no negative rebase;
2. **Liability Monotonicity:** `finalize()` increments atomically; `claim()` decrements atomically;
3. **Ownerless Distributor:** No administrative sweep or drain functions.

10 Appendix C: On-Chain Gas Profiling Benchmark

Operation / Call	Caller	Complexity	Measured Gas Units
<code>stake()</code> (initial)	User	$O(1)$	142,500
<code>stake()</code> (top-up)	User	$O(1)$	98,200
<code>redeem()</code> (full)	User	$O(1)$	125,600
<code>changeDiet()</code>	User	$O(1)$	68,400
<code>advanceInterval()</code>	Keeper	$O(M)$	185,000 (5 assets)
<code>claim()</code> (1 batch)	User	$O(1)$	88,300

<code>claim()</code> (10 batches)	User	$O(K)$	164,800
<code>claimThrough()</code>	User	$O(K_{\text{bd}})$	195,000 (bounded)
<code>claimNative()</code>	User	$O(K)$	112,000 (1 batch)

11 Appendix D: Verified Contracts Schedule

Component	Network	Verification Role
FATCAT Token	BNB Chain	1B fixed supply, immutable clone
PancakeSwap V2 Pool	BNB Chain	Locked liquidity, TWAP price feed
The Belly Vault	BNB Chain	Unprivileged exponential damped reservoir
StakingVault	BNB Chain	Isolated staking, unpausable redeem
SeniorityLedger	BNB Chain	$O(1)$ closed-form algebraic ledger
RewardAssetRegistry	BNB Chain	MENU registry with 5% probation cap
ExecutionRouter	BNB Chain	Delayed activation, MEV-protected router
RewardDistributor	BNB Chain	Dual-liability independent accounting
SeniorityCertificate (ERC-721)	BNB Chain	Post-launch roadmap; minted on exit burning FATCAT, 100% on-chain SVG
PancakeV2TwapOracle	BNB Chain	Endogenous TWAP price reader

12 Appendix E: Responsible Disclosure & Security Channel

1. **Official Security Channel:** Submit reports via GitHub Private Security Advisories;
2. **PGP Encryption:** Disclosures containing exploit proofs may be encrypted with the official PGP key;
3. **Review & Feedback:** Regular review with prompt triage feedback upon technical evaluation;
4. **White-Hat Grants:** Critical and High vulnerabilities receive substantial decentralized grants from the community treasury upon verification.

— Mathematics Belong to Everyone, Code is Law —